

Resolution Enhancement Technique

- 193 nm wavelength features.
- where we do photolithography.

features

optical proximity
correction

Double multi patterning



feature



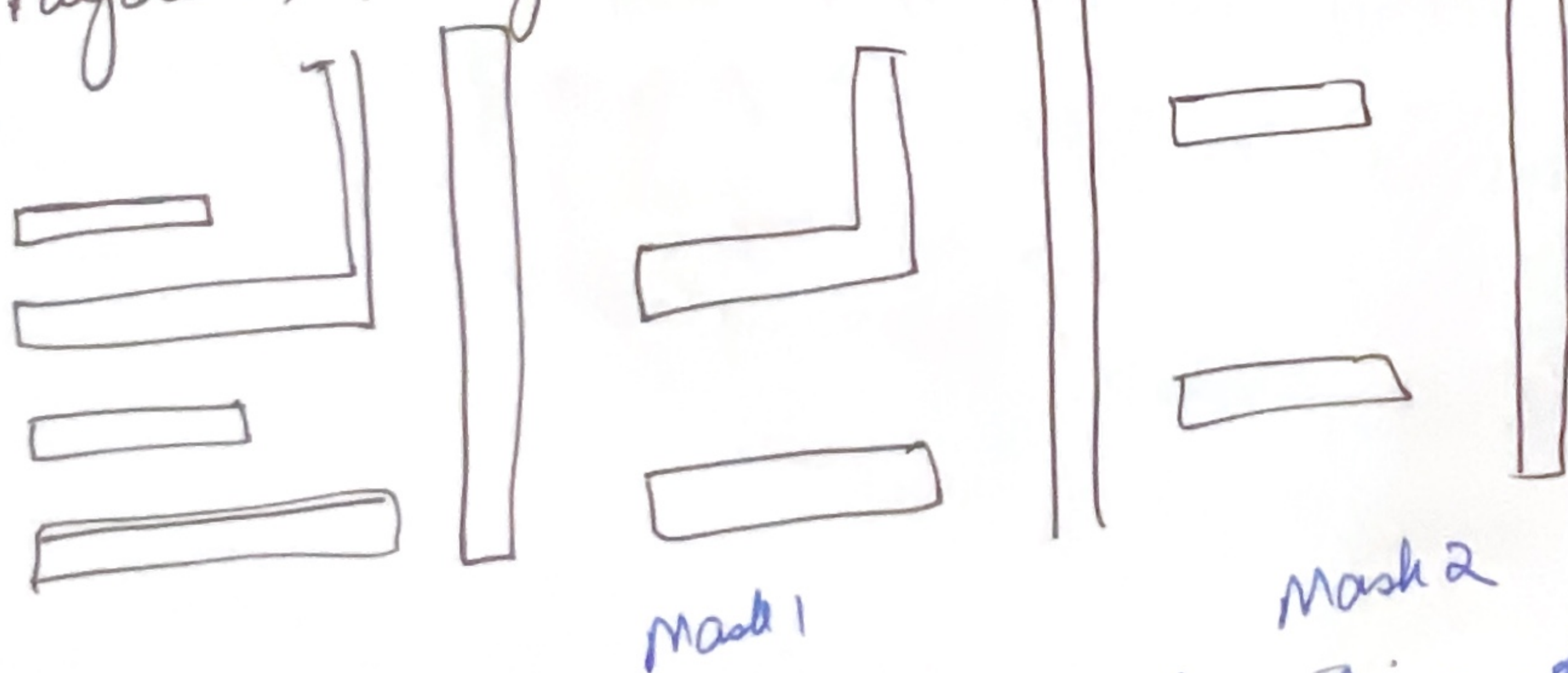
same
feature
on mask.

→ add features such as hammer head, serifs, mouse bites, To the mask. then, further distortion happens we get the same shape.

OPC

Double / Multi Patterning → if features are close, then they may overlap,

→ we decompose a closely spaced layout into 2 or more layouts, (assigned colors to features).



Wafer Fabrication

- Die with features
- Testing
- Packaging

→ functions of a Package

↳ Package provide pins for

connecting to external environ-
ment.

- have great impact, on the delay of the signal entering.

Heat dissipation.

Dual In line Package

Ball grid array (BGA)

Packages → and then final testing.

- high Voltage / high temperature
→ Burn in testing.

Binning

- Classification based on characteristics such as P_{max} , Power dissipation.

last → To change element in a test.

[upr $T - \$elem$]

↑ first evaluated

finished / Tested chips

TCL: Tool Command Language

set list { 0 1 2 3 4 }

set index -1

for each elem \$list

- HDL - Concurrency (Behavior) instructions → CPU →
→ Notion of time.

→ Some way, in case of Multiple Drivers

- Bit true data types, → A group in buses

⇒ for Purpose of Verification
→ Simulation

Verilog

Verification

logic

→ (VHDL)

Synthesized using Design Compiler.

2009

system Verilog.

Prefer ? when High impedance is dont care.

Data Types in Verilog - 0, 1, Z

net { wire, supply0, supply1, wand, wor.

Variables - store value in simulation
reg

for hex, octal, binary constants, x/x represents the unknown / dont care value, and z/z represents the high impedance

Value ? -> high impedance is dont care

case x x is dont care

case z ~~z is dont care~~

↳ Z, ? as dont care.

wire [31:0] data bus;
reg [7:0] address bus

difference b/w Z, ?

↑ don't care

8'b11 -> 00000011
8'z1 -> z z z z z z z 1

00000110
1111001

2's Comp.

string character - ASCII -> equivalent
Modules.

Ports allows communication b/w modules,

Parameter, default value can be overwritten.

#(
Parameter width = 16,
Parameter data = 0

)

is also negedge

Posedge -> 0, x, z -> 1
or 0 -> x, z

Negedge

-> 0, x, z

x, z, 1 -> 0
or x, z, 1
1 -> x, z

function	Tasks
function end fcn - Can have 1 or multiple i/p $\rightarrow$ 1 o/p	o / multiple Inputs & o / multiple o/p
Combinational ckt.	sequential ckt Modelling
Can call other fcn but Not Task.	can call tasks & functions
<u>No delay element</u>	<u>Delay element can be there</u>

LHS is Not assigned.

$\leq$ 2 step execution.

System Task and functions
 starts with $\rightarrow \$ \rightarrow$

- functional Verification

- L simulation $\rightarrow$ To verify that the o/p is in agreement
- L with the desired functionality for the given test stimuli.